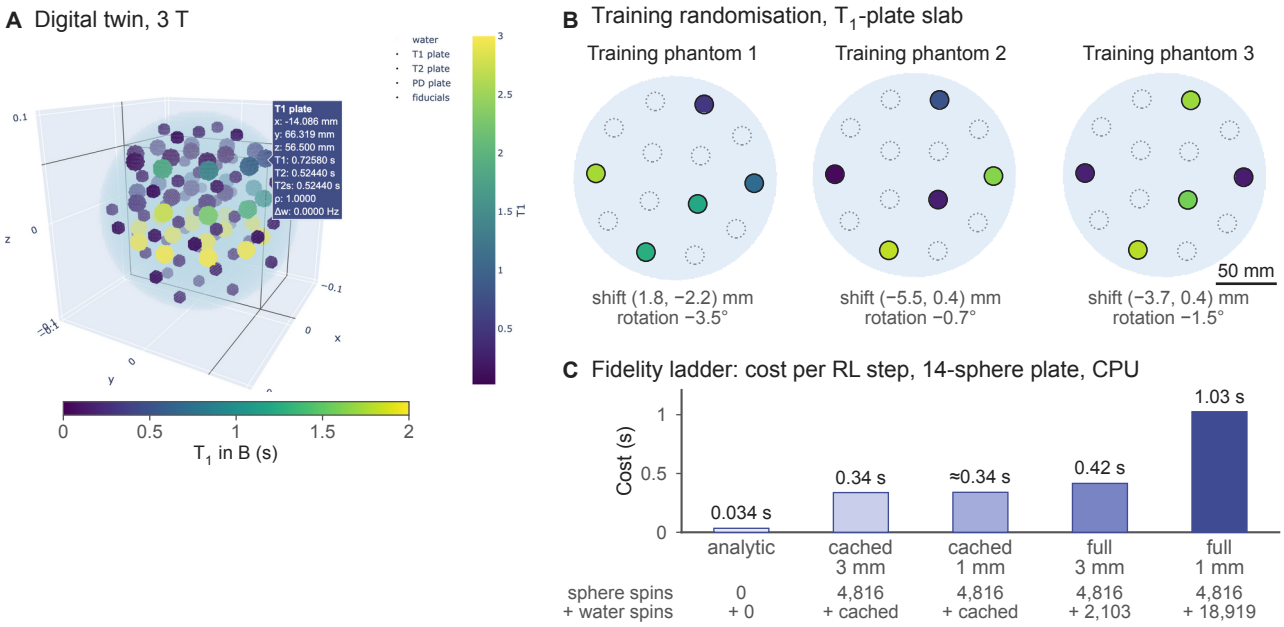

A Digital twin, 3 T
water
T1 plate
T2 plate
PD plate
fiducials
T1 plate
x: -14.086 mm
y: 66.319 mm
z: 56.500 mm
T1: 0.72580 s
T2: 0.52440 s
T2s: 0.52440 s
ρ: 1.0000
Δw: 0.0000 Hz
T1
T1 in B (s)
B Training randomisation, T1-plate slab
Training phantom 1
Training phantom 2
Training phantom 3
50 mm
shift (1.8, −2.2) mm
rotation −3.5°
shift (−5.5, 0.4) mm
rotation −0.7°
shift (−3.7, 0.4) mm
rotation −1.5°
C Fidelity ladder: cost per RL step, 14-sphere plate, CPU
Cost (s)
0.034 s
0.34 s
≈0.34 s
0.42 s
1.03 s
analytic
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm
sphere spins
+ water spins
0 + 0
4,816 + cached
4,816 + cached
4,816 + 2,103
4,816 + 18,919